Dual Axis Solar Tracker with Grid-Support

Sponsor: Student Self Led

Background:

The need for solar energy is increasing throughout the utility grid as well as efficiency in generation with the consideration of maintaining physical asset safety. Fixed solar panels suffer from gaining substantial energy losses throughout the day due to a non optimal mechanical motor that adjusts its panel board to the direct angle of the sun. Utility scale projects must take into consideration panel shading and mechanical wind damages that could affect the equipment. This project is looking to create a dual axis solar tracking platform that is integrated with dynamic grid support functions, high wind protection, and a monitoring system for the electric utility space.

Project Needs:

1). A motorized dual-axis tracking system to orient a solar panel towards the perpendicular light. PID controllers will be needed to calculate the maximum efficiency of changing angle.

2). Improvements to embedded control systems to track wind and storm routines. Also need PID controllers to help with anti shading logic.

3). Implementation of power measurement sensors and low voltage grid management to demonstrate energy arbitrage/

4). A created website dashboard for all data that is being collected in live time for users to be able to observe.

Project Scope:

This project encompasses design, prototyping, and web development for a dual axis solar tracking system integrated with intelligent protection controls. The mechanical side of the project includes the motor drive circuits, sensor, and DC voltage/current monitoring modules. On the software and controls side, it incorporates a tracking algorithm that aligns with the angle of incident of sun rays. The dashboard visualize real time tracking angles, generation curves, and simulated, along with control overrides for system benchmarks.

Potential Technical Considerations

Web Frontend:

- Python, Matlab

Control Backend:

- Microcontroller Designs, PID controllers,

Hardware and Physical Implementation:

- 3D Printing/CAD modeling, breadboard prototypes, DC Circuit Analysis

Deliverables:
1). Functional scale dual axis motorized solar tracking platform
2). Live operating web dashboard providing data metrics, control override systems, and visualizations of operation that is occurring.
3). Validation report delivering energy improvements with the use of a dual axis tracker and fixed-tilt reference panel.

Assumptions and Dependencies:
TBD